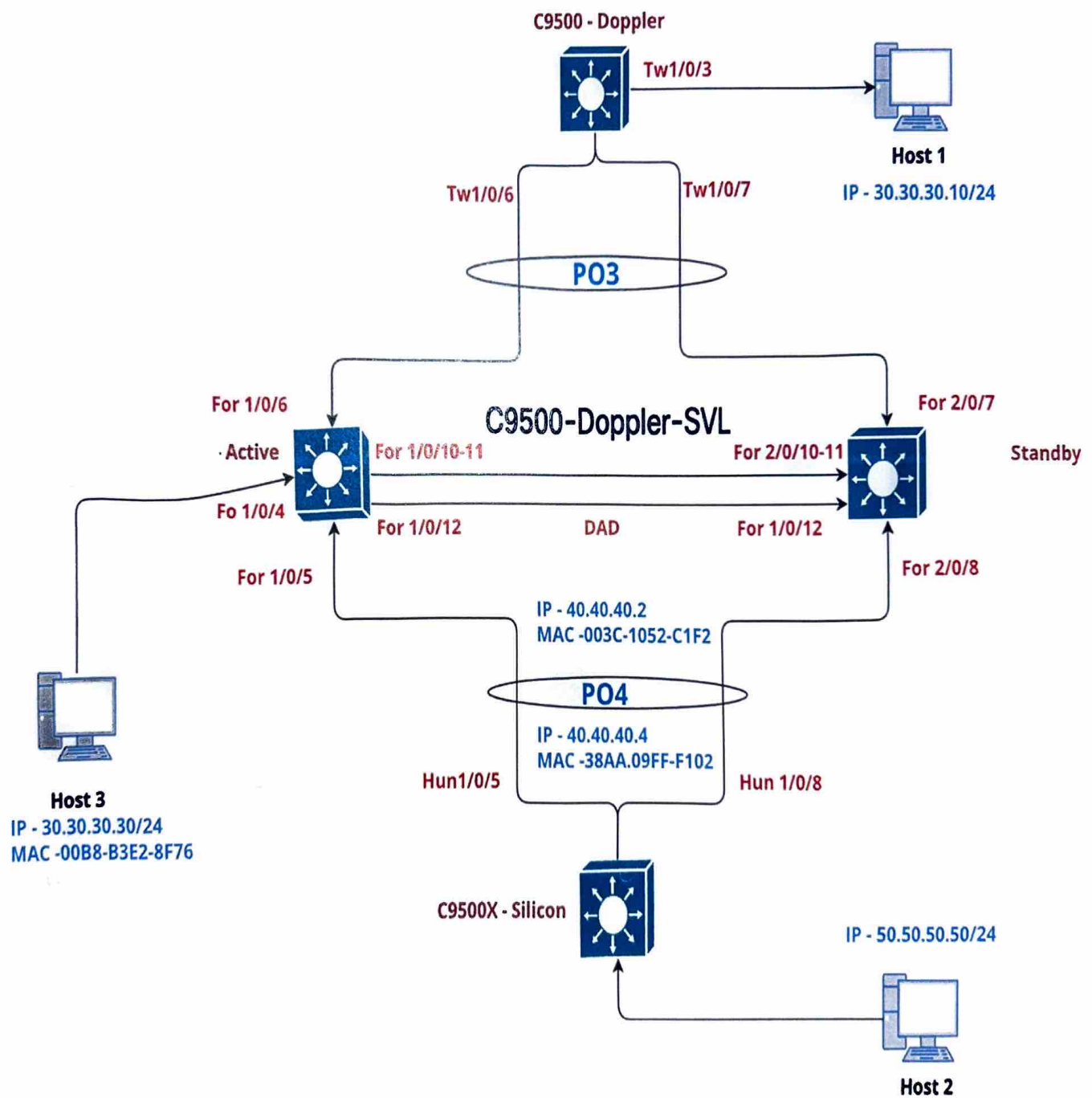
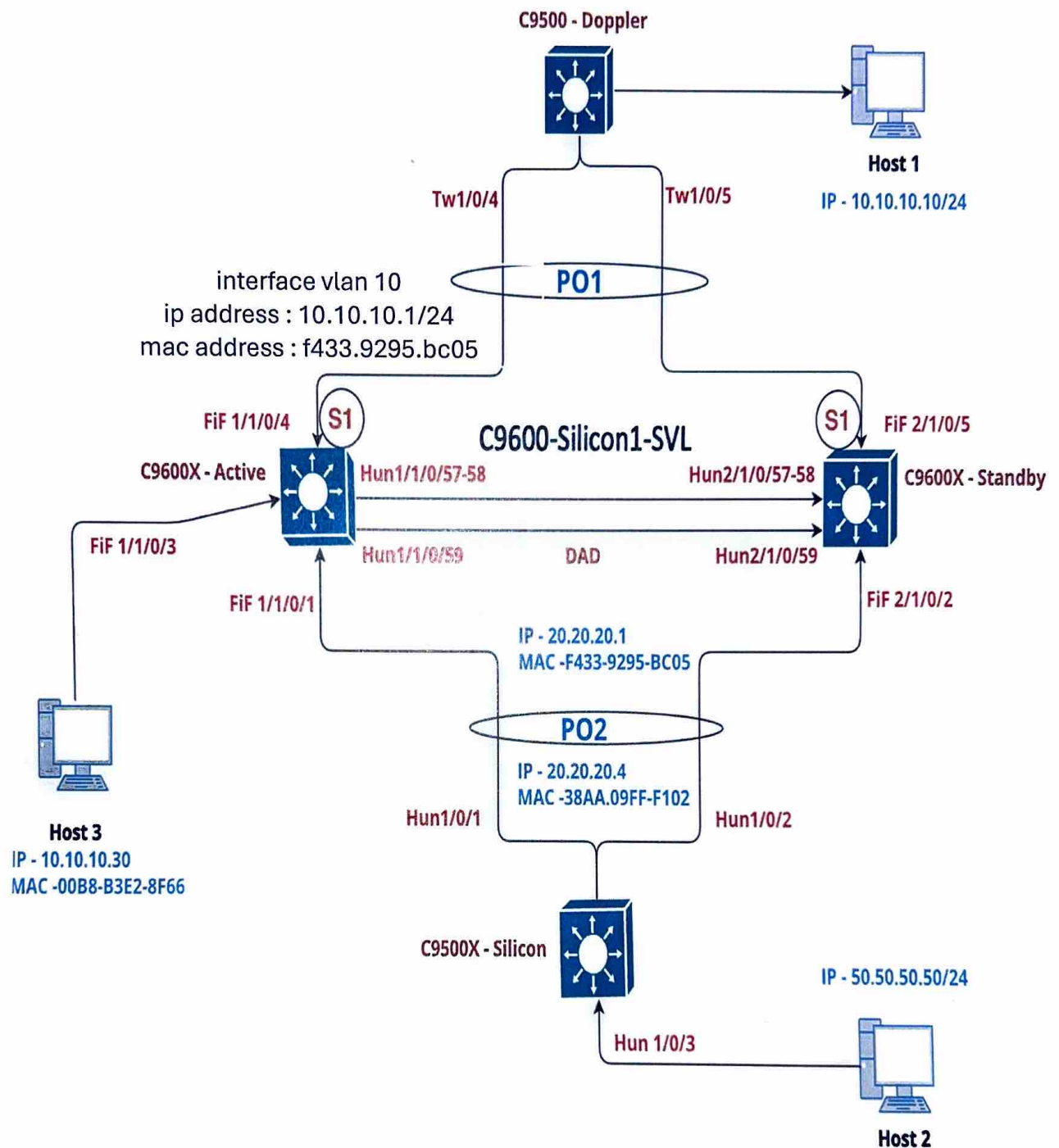


C9500 Doppler with Stackwise Virtual Link



C9600X Silicon ONE with Stackwise Virtual Link



Mastering Cisco Catalyst 9000 Switches - Layer 2 and Layer 3

Programming of Doppler Hardware Forwarding

Port Programming :-

```
show run interface fortyGigabitEthernet 1/0/4 + show interfaces
show platform software interface switch active R0 + F0 brief + name fort 1/0/4
show platform software fed switch active ifm mappings + if-id 12
show platform hardware fed switch active fwd-asic abstraction print-resource-handle ?
```

Vlan & STP programming :-

```
show vlan brief + show spanning-tree vlan 30
show platform software vp switch active R0 + F0 summary
show platform software fed switch active vp key 12 30 + vlan + vlan 30
show platform hardware fed switch active fwd-asic abstraction print-resource-handle
```

MAC Programming :-

```
show mac address-table interface fortyGigabitEthernet 1/0/4
show mac address-table address 00b8.b3e2.8f76
show platform software matm switch active R0 + F0 mac 00b8.b3e2.8f76
show platform software object-manager switch active F0 object 3945 | parents
show platform software fed switch active matm macTable vlan 30 mac 00b8.b3e2.8f76
show platform hardware fed switch active fwd-asic abstraction print-resource-handle ?
show plat hard fed switch active fwd-asic resource asic all destination-index range <>
```

L3 Programming :-

```
show ip route 50.50.50.50 + show ip cef 50.50.50.50
show ip cef exact-route 30.30.30.30 50.50.50.50
show platform software ip switch active R0 | F0 cef prefix 50.50.50.0/24
show platform software adjacency switch active F0 index 0x2b
show ip arp | i f102|c1f2
show platform software fed switch active ip route 50.50.50.0/24 + detail
show platform hardware fed switch active fwd-asic abstraction print-resource-handle
0x733690847f28 1
show platform hardware fed switch active fwd-asic resource asic all rewrite-index range
0x15 0x15
show ip arp | i f102
show platform hardware fed switch active fwd-asic resource asic all destination-index
range 0x4c9f 0x4c9f
```

Programming of Silicon ONE Hardware Forwarding

L2 Forwarding :-

```
show run int Fif1/1/0/3 || show interfaces Fif1/1/0/3 + switchport + trunk
show idprom interface FiftyGigE1/1/0/3
show vlan brief + show spanning-tree vlan 10
show mac address-table int Fif 1/1/0/3 + show mac address-table add 00b8.b3e2.8f66
show platform software matm switch active R0 + F0 mac 00b8.b3e2.8f66
show platform software object-manager switch active F0 object 14077 + Parents
show plat soft fed switch active matm MacTable vlan 10 mac 00b8.b3e2.8f66 detail
show platform software fed switch active vp key 1035 10 + details
show platform software fed switch active vlan + vlan 10
show platform hardware fed switch active fwd-asic insight sdk_object(1938) + (0x813)
show platform software fed switch active ifm mappings
show platform software fed switch active ifm interface_name FiftyGigE1/1/0/3
show platform hardware fed switch active fwd-asic insight sdk_object(751) + (738)
```

L3 Interface Programming :- show run interface vlan 10 + show interfaces vlan 10

```
show platform software interface switch active R0 | F0 name vlan 10
show platform software object-manager switch active F0 Object 439
show platform software fed switch active ifm interfaces svi + interface_name vlan 10
show platform hardware fed switch active fwd-asic insight sdk_object(2071)
```

L3 Directly Connected Routes :- show ip route 10.10.10.30

```
show ip cef 10.10.10.30 + show ip cef exact-route 10.10.10.1 10.10.10.30
show platform software ip switch active R0 cef prefix 10.10.10.30/32
show platform software ip switch active R0 cef prefix 10.10.10.30/32 detail
show platform software adjacency switch active R0 index 18 <ARP check>
show platform software ip switch active F0 cef prefix 10.10.10.30/32
show platform software ip switch active F0 cef prefix 10.10.10.30/32 detail
show platform software adjacency switch active F0 index 0x12
show platform software object-manager switch active F0 object 6889
show platform software object-manager switch active F0 object 6889 parents
show platform software fed switch active ip route 10.10.10.30/32
show platform software fed switch active ip route 10.10.10.30/32 detail
show platform hardware fed switch active fwd-asic insight sdk_object(0x817)
```

L3 Indirect Route : via Routing Protocol :- show ip route 50.50.50.50

```
show ip cef 50.50.50.50 + show ip cef exact-route 10.10.10.10 50.50.50.50
show platform software ip switch active R0 + F0 cef prefix 50.50.50.0/24
show platform software adjacency switch active R0 + F0 index 0x24
show platform software ip switch active F0 cef prefix 50.50.50.0/24 detail
show platform software object-manager switch active F0 object 6543 | parents
show platform software fed switch active ifm mappings etherchannel
show platform software fed switch active ip route 50.50.50.0/24 | detail
show platform hardware fed switch active fwd-asic insight sdk_object(0x6cf)
show platform hardware fed switch active fwd-asic insight sdk_object(0x820)
```